

Fp Growth Algorithm

FP Growth Algorithm

- FP-Growth algorithm is also used for frequent pattern mining.
- **The FP-Growth or Frequent Pattern Growth algorithm is an advancement to the apriori algorithm.** While using the apriori algorithm for association rule mining, we need to scan the transaction dataset multiple times. In the FP growth algorithm, we just need to scan the dataset twice.
- **An FP-Tree is a tree data structure created from the transaction data while generating frequent itemsets in the FP growth algorithm.**
- To create an FP-Tree, we first scan the transaction dataset and record the support count of each item. Then, we create a tree structure where each node in the tree represents an item in the dataset and its frequency count.
- The root node has no associated item and is used as a starting point for the tree. We denote the root node by None or Null. The children of a node in the fp-tree represent the items that frequently co-occur with the parent item in the dataset.

FP Growth Algorithm-Example

Transaction ID	Items
T1	I1, I3, I4
T2	I2, I3, I5, I6
T3	I1, I2, I3, I5
T4	I2, I5
T5	I1, I3, I5

FP Growth Algorithm-Example

- Step 1: Calculate The Support Count of Each Item in The Dataset

Item	Support Count
I1	3
I2	3
I3	4
I4	1
I5	4
I6	1

Support Count of Items in the dataset

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- let's sort the above table by the support count

Item	Support Count
I3	4
I5	4
I1	3
I2	3
I4	1
I6	1

Sorted Support Count of Items in the Dataset

FP Growth Algorithm-Example

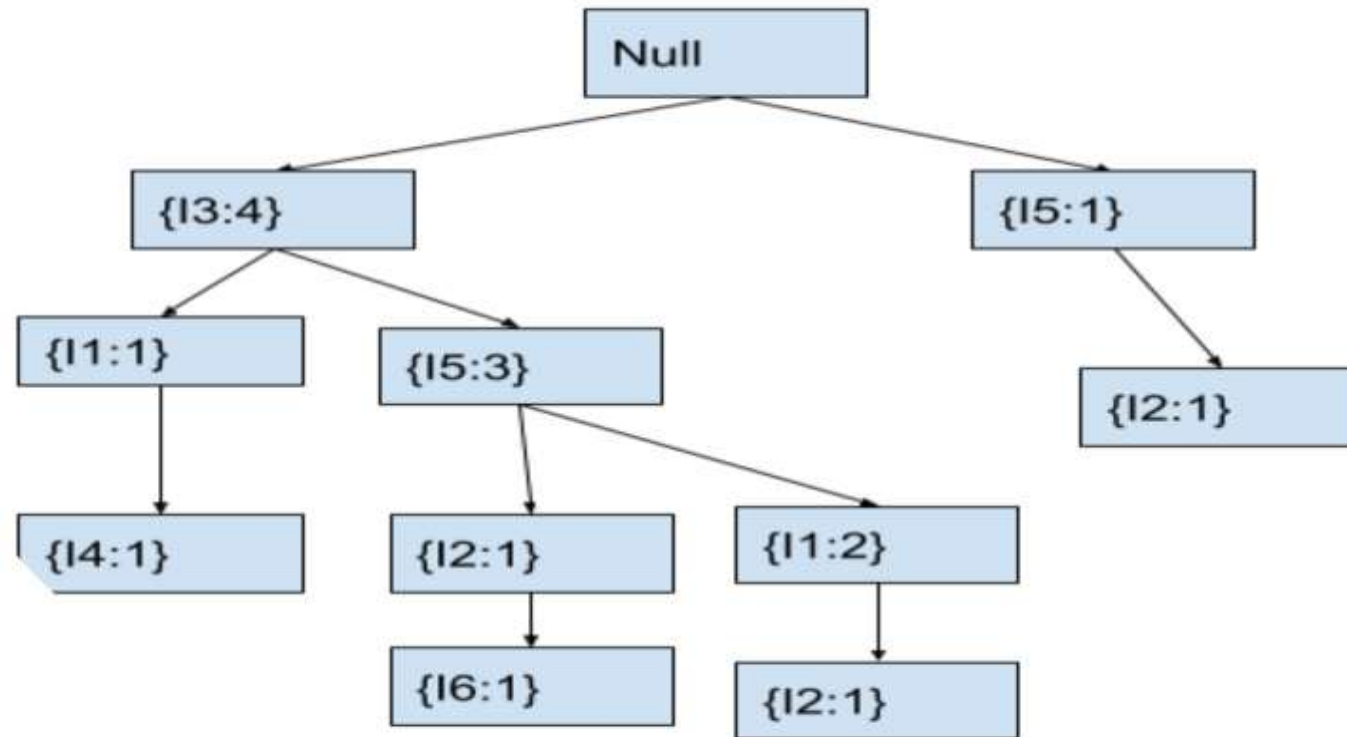
- Step 2: Reorganize The Items in The Transaction Dataset

Transaction ID	Items
T1	I3, I1, I4
T2	I3, I5, I2, I6
T3	I3, I5, I1, I2
T4	I5, I2
T5	I3, I5, I1

Dataset with sorted items in the transactions

FP Growth Algorithm-Example

- Step 3: Create FP Tree Using the Transaction Dataset



Final FP-Tree for the FP Algorithm Numerical Example

FP Growth Algorithm-Example

- Step 4: Create a Pattern Base For All The Items Using The FP-Tree

The pattern base for each item in the dataset is tabulated below.

Item	Pattern Base
I1	{I3}:1,{I3, I5}:2
I2	{I3, I5}:1,{I3, I5, I1}:1,{I5}:1
I3	{}
I5	{I3}:3

Pattern Base created in FP Growth Numerical Example

FP Growth Algorithm-Example

- **Step 5: Create a Conditional FP Tree For Each Frequent Item**

The conditional fp-tree for each item is tabulated below.

Item	Pattern Base	Conditional FP Tree
I1	{I3}:1,{I3, I5}:2	{I3:3, I5:2}
I2	{I3, I5}:1,{I3, I5, I1}:1,{I5}:1	{I3:2, I5:3}
I3	{}	
I5	{I3}:3	{I3:3}

Conditional FP-Tree for items

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- **Step 6: Generate Frequent Itemsets From Conditional FP-Trees**

All the frequent itemsets derived from the conditional fp-trees have been tabulated below.

Item	Pattern Base	Conditional FP Tree	Frequent itemsets
I1	{I3}:1,{I3, I5}:2	{I3:3, I5:2}	{I1, I3}:3,{I1, I5}:2,{I1, I3, I5}:2
I2	{I3, I5}:1,{I3, I5, I1}:1,{I5}:1	{I3:2, I5:3}	{I2, I3}:2, {I2, I5}:3,{I2, I3, I5}:2
I5	{I3}:3	{I3:3}	{I3, I5}:3

Frequent itemsets generated in the FP Growth Numerical example